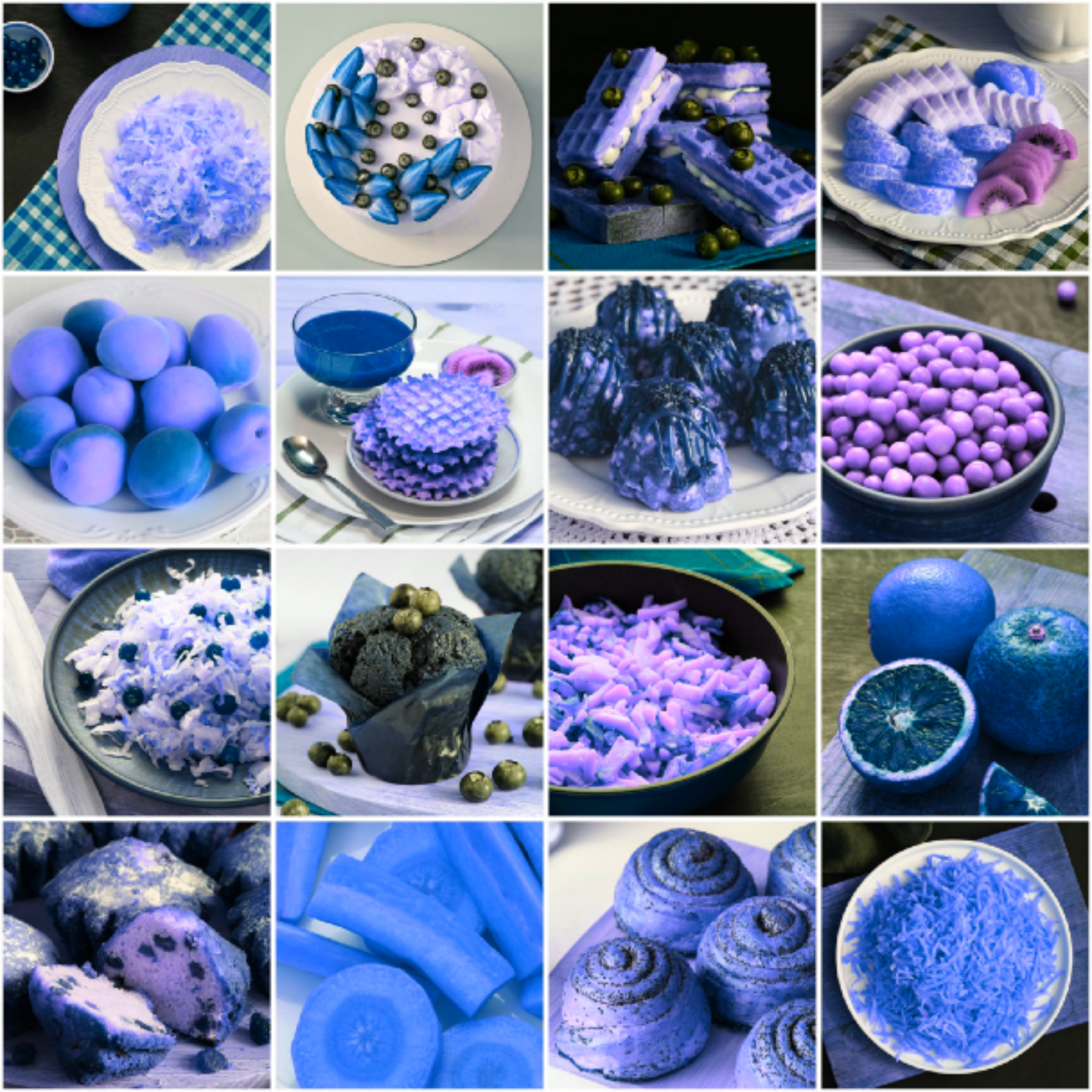
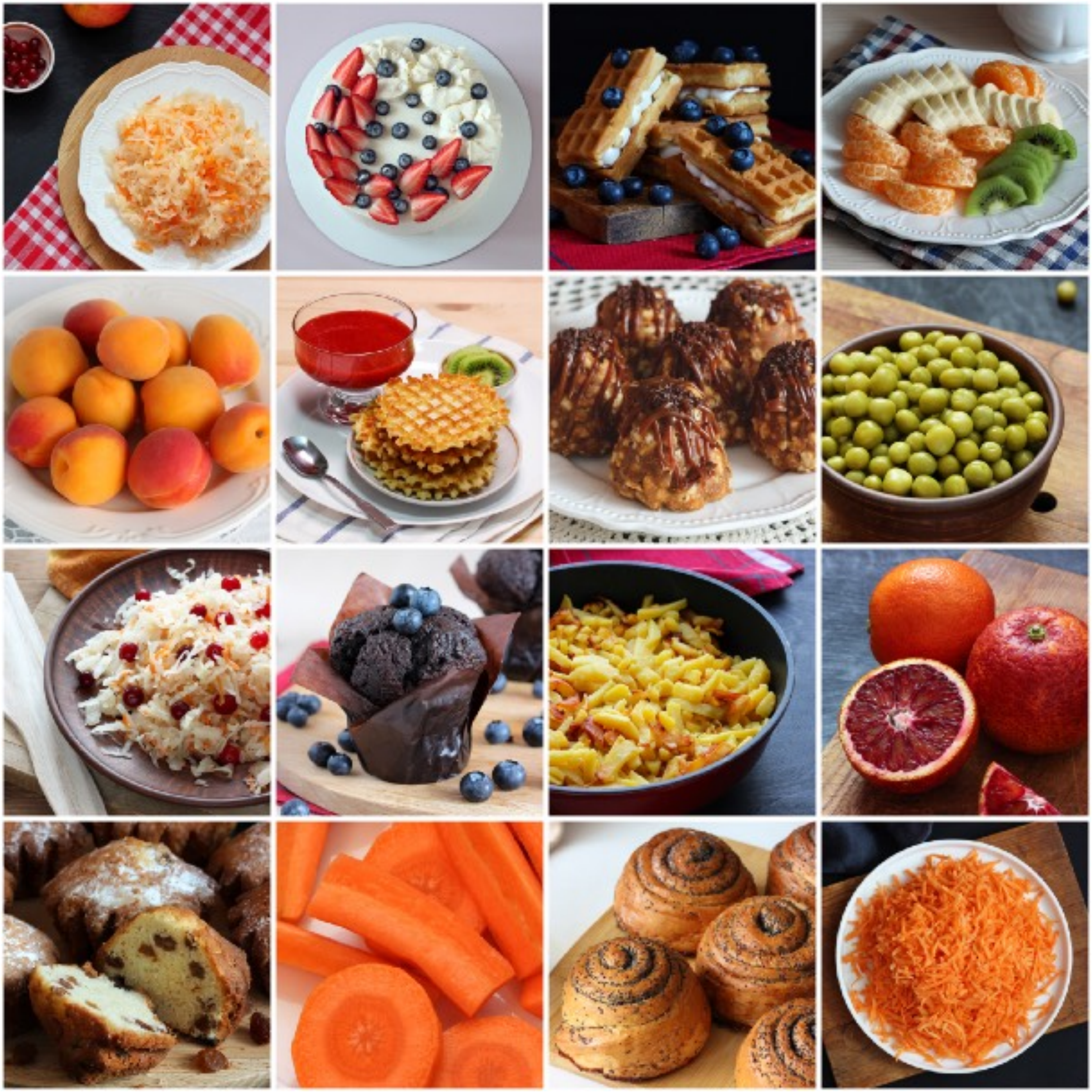
SUPERVISED LEARNING

What Kind of Data Types?





The dataset is a collection of tuples:

$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$ Where:

- $x^{(i)} = [x_1^{(i)}, x_2^{(i)}, \ldots, x_m^{(i)}]$ is the numerical representation of the image for classification. It is created by flattening pixel values into a 1D array. For instance, a 100x100 pixel image yields a 10,000-element grayscale vector or a 30,000-element RGB vector.
- $y =$ The probabilities of x for each class/category.

Example: Healthy vs. Unhealthy Food

Image Classification

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- $y =$ The bounding box of objects.